

MTE 204 – Numerical Methods

Tutorial #5 Question

Question 1:

Compute

$$\int_{0.5}^{2.5} \frac{1}{x^2} dx$$

Using

- a) Trapezoid rule with $n = 1, 2, 4$, and 8 segments
- b) Romberg integration with $n = 1, 2, 4$, and 8 segments
- c) Two-point Gauss quadrature

For each method, discuss convergence behaviour, the number of function evaluations, and the error of the integral approximation.